

Engineering Electronics – Diodes and Rectifiers

Diode

A diode is a semiconductor device that mainly allows current to flow in one direction. It consists of a p-type region and an n-type region forming a PN junction.

Forward Bias

In forward bias, the positive terminal of the supply is connected to the p-side and the negative terminal to the n-side. The depletion region becomes smaller and current can flow when the applied voltage is sufficient.

Reverse Bias

In reverse bias, the depletion region becomes wider and only a small leakage current normally flows. Excessive reverse voltage can cause breakdown.

Rectifier

A rectifier converts alternating current into pulsating direct current. Half-wave rectifiers use one diode, while full-wave rectifiers use both halves of the AC waveform.

Bridge Rectifier

A bridge rectifier commonly uses four diodes. During each half cycle, two diodes conduct so that the load current maintains the same direction.

Troubleshooting

If a rectifier produces no output, check the input supply, diode connections, diode condition, fuse, transformer, and load. A shorted or open diode can cause abnormal output.